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The unfinished business of evidence strength in software engineering: Current practices and future directions

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Abstract

Context: Almost 20 years ago, the Software Engineering research community was introduced to the concept of evidence strength: the extent of confidence that the estimates of the effect of an intervention are correct. The motivation for assessing the strength of evidence is very practical: if one has high confidence in the beneficial effects of an intervention and its low harm, one can make strong recommendations for its adoption in the daily work of practitioners. Since then, software engineering researchers have started using GRADE (Grades of Recommendation, Assessment, Development, and Evaluation) as an approach to the assessment of strength of evidence. However, to the present day, no study has evaluated the extent to which systematic literature reviews in software engineering have effectively assessed the strength of evidence supporting their review findings.

Objective: We aim to understand the current strength assessment of evidence in software engineering secondary studies, answering questions about the most widely used methods, how researchers use or adapt them, and whether software engineering researchers make recommendations to practitioners from their review findings.

Method: We executed a tertiary study using automated and snowballing search procedures to accomplish our goal. For data extraction and analysis, we selected 22 papers in which the authors claimed to perform a strength assessment of evidence.

Results: GRADE is the most used method, although not necessarily appropriate considering the type of review. Additionally, software engineering researchers are not benefiting from its more recent versions or GRADE-CERQual (Confidence in the Evidence from Reviews of Qualitative research), a method specific to qualitative reviews. We also found that most strength assessments are generic for all review findings instead of specific for each review finding or outcome of interest, as the GRADE and GRADE-CERQual approaches recommend. Furthermore, very few reviews make explicit recommendations to practitioners based on their review findings, and none of them evaluated the strength of such recommendations.

Conclusions: A fundamental issue our study brings forward is that if the Software Engineering community decides to apply the original methods "as is" there is plenty of room for improvement in their faithful adoption. However, more than just adopting existing

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frameworks blindly—and at times incorrectly—it is important to reflect on what kinds of evidence our community produces and how best to assess it given the particularities of our field. Adapting existing methods could represent a step forward, yet SE researchers might also need to consider developing new ones from scratch.

Keywords Strength of evidence · Qualitative research · Systematic literature reviews · Software engineering

1 Introduction

The paradigm of Evidence-based Software Engineering (EBSE) emerged 20 years ago, inspired by Evidence-Based Medicine (EBM) (Kitchenham et al. 2004). Evidence-Based Medicine is a movement that aims to improve medical practice by integrating clinical expertise, patients' values/preferences, and the best available evidence (Chloros et al. 2023). Similarly, the goal of EBSE is "to provide the means by which current best evidence from research can be integrated with practical experience and human values in the decision-making process regarding the development and maintenance of software" (Kitchenham et al. 2004). In other words, researchers adopt EBSE to produce evidence about whether relevant practices lead to real improvements, and practitioners base their daily decisions on developing and maintaining software on such evidence.

The novelty Evidence-Based Medicine introduced was not the use of scientific evidence in practice per se, but rather (1) defining what constitutes the best evidence and (2) establishing how to gather it (Feinstein and Horwitz 1997). Regarding the first point, for a long time, the best evidence was defined by a simple hierarchy, typically represented by a pyramid based on methodological grounds (Mercuri and Baigrie 2019). At the top of this pyramid sit randomized controlled trials, while observational studies and expert opinion reside at the bottom. As for the second point, one of Evidence-Based Medicine's core principles is evaluating the totality of evidence in the pursuit of truth, rather than selecting pieces that support a particular claim (Djulbegovic and Guyatt 2017). This principle is implemented through Systematic Literature Reviews (Djulbegovic and Guyatt 2017), a method the SE research community has embraced.

A few years after the paper by Kitchenham et al. (2004), Dybå and Dingsøyr (2008a) expressed concerns about how much confidence one can place on conclusions and recommendations of Software Engineering (SE) systematic literature reviews. The authors discussed two issues to address these concerns. The first regards assessing the quality of the individual primary studies included. Yang et al. (2021) showed an upward trend, starting in 2007, of systematic literature reviews applying quality assessment instruments to evaluate these studies, indicating the SE research community is tackling this first issue.

The second concern in dealing with confidence is assessing the strength of the evidence of systematic literature review findings (Dybå and Dingsøyr 2008a). EBM's hierarchies of evidence attempt to structure such assessments but have been heavily criticized for their oversimplification (Djulbegovic and Guyatt 2017; Chloros et al. 2023). These hierarchies stratify evidence based on study design (Chloros et al. 2023) and focus almost exclusively on randomized controlled trials (Martini 2021). However, such trials are sometimes infeasible or unethical (Chloros et al. 2023; Feinstein and Horwitz 1997) and cannot answer all relevant

research questions (Chloros et al. 2023). Moreover, some of the most conclusive medical evidence has come from observational studies, such as those on the benefits of insulin or the harms of smoking (Chloros et al. 2023). These criticisms resonate with the reality of SE research, where randomized controlled trials are rare (Kitchenham et al. 2015, p. 327).

Over time, a more sophisticated alternative to simple pyramid-based hierarchies emerged: the GRADE (Grades of Recommendation, Assessment, Development, and Evaluation) methodology (Guyatt et al. 2011a; Dybå and Dingsøyr 2008a). It incorporates factors beyond study design to assess the strength of evidence. It also allows upgrading the strength of evidence from observational studies, which might be particularly useful in domains like SE, where randomized trials are uncommon. Additionally, GRADE-CERQual (Confidence in the Evidence from Reviews of Qualitative Research) was developed to support qualitative systematic reviews (Lewin et al. 2018a).

Recently, Kitchenham et al. (2022) discussed the use of GRADE and GRADE-CERQual in SE secondary studies. However, since Kitchenham et al. (2022) focus on an overall checklist for reporting systematic reviews, they do not provide a comprehensive view of the current state of SE systematic reviews regarding this topic. For perspective, we consider a tertiary study by Budgen et al. (2020) on how much support systematic reviews provide for teaching SE practice. This study found that very few reviews assessed the strength of the evidence supporting their findings. This hints that we, as a research community, are discussing strength assessment without necessarily conducting it.

In this study, we present a tertiary study that addresses the current lack of knowledge within the SE community on how researchers assess the strength of evidence in their findings—particularly given the time elapsed since the work of Dybå and Dingsøyr (2008a). Our results reveal substantial room for improvement. For instance, among 74 SE systematic literature reviews citing Dybå and Dingsøyr (2008a) out of 94 papers that passed the first filter of selection, only twelve attempted to assess the strength of supporting evidence. This suggests that the SE research community is making very limited use of such assessments. More importantly, it indicates that we are missing an opportunity to engage in deeper discussions about how to assess the strength of bodies of evidence in our field–especially since no other papers since Dybå and Dingsøyr (2008a) have critically examined this topic. We believe it is timely to reflect on whether the SE community should adopt or adapt established methods from EBM, or even develop its own tailored guidelines.

Any decision to adopt or adapt GRADE or GRADE-CERQual should be made with a clear understanding of their strengths and weaknesses. On the one hand, both frameworks have been in development for many years in other fields and can serve as useful starting points for SE. They include explicit components for assessing evidence strength, which have the potential to make such judgments more systematic, accountable, and transparent (Balshem et al. 2011). Moreover, empirical evidence suggests that appropriate training in GRADE can improve consistency and agreement among different raters (Dorneles et al. 2024).

On the other hand, several weaknesses have been identified. Critics have noted the limited theoretical and empirical justification for some of the specific components proposed for strength assessments (Mercuri and Gafni 2018a, b, c). Others have pointed to the vagueness of certain terminology (Mercuri et al. 2018), and to the considerable time and resource demands of applying these methods in practice (Wyer 2018). This is not an exhaustive list, but it does highlight the need for the SE community to reflect carefully on whether adopting these methods "as is" is the best path forward. Our intention is not

to advocate for GRADE or GRADE-CERQual uncritically, but rather to present them as possible options–among others–to be weighed and debated. The decision on whether, and how, to use them should ultimately be guided by their fit to the research questions, contexts, and methodological traditions of SE.

Additionally, GRADE outlines concrete steps for translating evidence into practitioner guidelines (Guerra-Farfan et al. 2023). Panels of experts carry out this task, building on systematic review findings under the responsibility of medical societies and organizations. These panels use GRADE to create clinical practice guidelines on a wide range of topics—from symptom management (e.g., of adults with serious respiratory illness by the European Respiratory Society; Holland et al. 2024) to supplement recommendations (e.g., vitamin D by the Endocrine Society; Demay et al. 2024) and pharmacological treatments (e.g., for obesity by the American Gastroenterological Association; Grunvald et al. 2022).

Nevertheless, this approach to practice has led to criticism of Evidence-Based Medicine for encouraging a "cookbook" approach (Martini 2021)—potentially dogmatic, rigid, and too uniform (Siwek 2018). This criticism often stems from concerns about treating individuals who differ from the average patient in randomized trials (Djulbegovic and Guyatt 2017), which could put minorities at risk (Stevens 2018). Others argue it favors rules over clinical judgment (Djulbegovic and Guyatt 2017). However, Evidence-Based Medicine supporters assert that it does not exclude judgment but requires *judicious judgment*, incorporating individual values and preferences (Djulbegovic and Guyatt 2017).

Moreover, GRADE requires expert panels to consider the best existing evidence along with relevant information from the practitioner's[1] perspective, including their values and preferences, the balance of benefits and harms, and resource requirements (Andrews et al. 2013). This has the potential to support SE practitioners in making timely decisions about adopting methods, practices, or tools, since all essential information is summarized in expert-issued guidelines. Ultimately, the SE community can leverage GRADE and GRADE-CERQual to help bridge the gap between industry and academia.

Therefore, this study aims to provide a critical discussion on assessments of strength of evidence in Software Engineering. We hope this will reignite reflection on this topic within our community, serving as a next step towards the full implementation of evidence-based principles in SE, just as it was a fundamental step in Evidence- Based Medicine. We strongly believe that the consistent application of EBSE principles, with explicit emphasis on evidence strength, is a key enabler for bridging the gap between research and industry practice. Therefore, this paper may be especially useful for:

- Authors of systematic reviews, as it will clarify what methods and elements to use for strength assessment;
- Reviewers of systematic reviews, as it will clarify what to look for in strength assessments;
- Researchers improving systematic review procedures, as it will identify misconceptions and potential adaptations of existing methods for SE;
- Leaders of associations or committees responsible for publishing bodies of knowledge or guidelines in SE, since strength assessment is key for evidence-based recommendations.

The rest of the paper presents:

[1] In the original: patient. We changed this to adapt to the Software Engineering context

- A thorough review of GRADE and GRADE-CERQual methodologies (Section 2);
- The study methodology (Section 3);
- A detailed account of their use in SE (Section 4);
- A discussion of their misuses and suggestions for improving strength assessment practices in SE systematic literature reviews (Section 5);
- Threats to validity (Section 6);
- Final considerations (Section 7).

Next, in Section 2, we present more about GRADE and GRADE-CERQual.

2 Background

This section provides an overview of GRADE and GRADE-CERQual, methods designed for quantitative and qualitative reviews, respectively. According to Kitchenham et al. (2022), a qualitative review is a secondary study in which authors use a qualitative method to synthesize findings of the primary study. Similarly, we define a quantitative review as one that uses a quantitative synthesis method[2].

Section 2.1 revisits GRADE, highlighting the importance of identifying outcomes of interest in quantitative systematic literature reviews and presenting the components of its most recent version. It also distinguishes between assessing *the strength of evidence* and *the strength of recommendations*. Section 2.2 introduces GRADE-CERQual, discussing how qualitative systematic literature reviews summarize review findings and the components used to assess the strength of the supporting evidence.

Our goal with this section is not to provide detailed guidance on each approach, as other papers extensively describe them, nor to advocate for their blind adoption. Instead, we aim to provide an overview of key concepts and ideas relevant to understanding the current state of GRADE and GRADE-CERQual's use in SE research and their potential to improve our practices.

2.1 Revisiting GRADE

GRADE is an approach to rate the quality of evidence and the strength of recommendations, focused on authors of systematic reviews, health technology assessments, and guideline developers (Guyatt et al. 2011a). The initiative started in 2000 and has evolved over the years. The first GRADE paper was published in 2004 (Atkins et al. 2004), marking the initial version of the method. An updated version followed in 2008 (Guyatt et al. 2008). The final series of papers explaining the approach (Guyatt et al. 2011a), which we explore in this section, has been published since 2011. When using GRADE, the rating of evidence occurs at the level of the outcomes of interest of the review, not at the level of each primary study (Guyatt et al. 2011b). This requires the synthesis of the primary studies' findings regarding each outcome of interest. The synthesis culminates with the development of Summary of Findings (SoF) Tables, presenting the assessment of the

[2]A well-known synthesis method for quantitative reviews is a statistical meta-analysis (Kitchenham et al. 2015, Chapter 3). However, quantitative reviews can also employ other synthesis methods, such as vote counting or combining p-values (Campbell et al. 2020)

quality of evidence for each outcome. It also results in Evidence Profile (EP) Tables, presenting an explicit evaluation of each GRADE component determining the quality of evidence for each outcome along with their corresponding Summary of Finding Tables (Guyatt et al. 2011c).

2.1.1 Research Questions and Outcomes of Interest

Research questions can help focus intervention assessments on the outcomes that can support practical decision-making—that is, on the variables practitioners will consider relevant to choose one intervetion over another. In Software Engineering, outcomes of interest may include quality and productivity, among others. GRADE suggests PICO (population, intervention, comparison, and outcomes) for the clear definition of research questions in systematic reviews (Guyatt et al. 2011b). This structure has been used for deriving search strings in Software Engineering systematic reviews, though it has not worked well for this purpose (see Kitchenham et al. 2022, Supplementary Material)."

Since many outcomes may be associated with an intervention, it is often not feasible to collect and assess evidence for all of them. Therefore, it is important to prioritize, distinguishing those that are critical, important, or of low relevance to decision-making (Guyatt et al. 2011b). In cases where evidence is lacking for a critical outcome, this gap should be acknowledged rather than overlooked. For infrequent outcomes, surrogates may be used. The next step is to gather and synthesize evidence related to each outcome and assess the effects of the intervention accordingly, considering the dimensions in Section 2.1.2.

2.1.2 GRADE Components

Considering that GRADE is meant for the authors of systematic reviews and/or of guidelines (such as clinical practice guidelines), its components do not apply to a single study but to a body of evidence supporting a conclusion about an outcome of interest. GRADE acknowledges that the quality of a body of evidence involves more than just the risk of bias in individual studies. The framework proposes that one must consider four additional components to judge confidence in estimates of intervention effects (Balshem et al. 2011), as Table 1 presents.

The assessment process begins with the assignment of an initial score based on the study design of the primary studies. Randomized controlled trials (RCTs) receive a high initial rating. Researchers then apply Table 1 components to decide whether to lower the confidence level. In contrast, GRADE requires assigning a low initial rating to observational studies. Researchers can raise the confidence level when specific criteria apply, as Table 2 summarizes.

Guyatt et al. (2011i) caution against raising confidence due to large effects, especially when outcomes are subjective or when there are serious concerns on other GRADE components, such as bias, imprecision, or publication bias.

After applying all relevant criteria, reviewers assign a confidence level to the effect estimate for each outcome, as Balshem et al. (2011) describe:

- **High:** We are very confident that the effect estimate is close to the true effect.
- **Moderate:** We are moderately confident in the effect estimate, as there is a possibility it is substantially different from the true effect.

Table 1 GRADE components for lowering scores

Components	Description
Risk of bias	Problems in study design or execution that can mislead results—also known as study limitations and internal validity issues (Guyatt et al. 2011d). In randomized controlled trials, this includes lack of anonymization, selective reporting, early stopping for benefit, and loss to follow-up* Guyatt et al. (2011d).
Publication bias	The tendency to withhold negative or non-significant results, leaving them unpublished or publishing them only in abstracts, theses, or low-visibility venues (Guyatt et al. 2011e).
Imprecision	Extent to which the confidence intervals (CIs) around the difference in effect between the intervention and the comparison are satisfactorily narrow (Guyatt et al. 2011f). Researchers should downgrade confidence when confidence intervals are so wide that different bounds would lead to different recommendations (Guyatt et al. 2011f).
Inconsistency	Unexplained variability in results across studies, even after examining potential causes. Researchers assess this based on how similar the effect estimates are, whether confidence intervals overlap, and results of statistical heterogeneity tests (Guyatt et al. 2011g).
Indirectness	Mismatch between the population, intervention, or outcomes studied and those targeted by the review or guideline (Guyatt et al. 2011h). Indirect comparisons, where studies compare each intervention with a third, rather than with each other, also introduce indirectness (Guyatt et al. 2011h).

* Loss to follow up refers to the loss of participants during the course of the study due to varied reasons and leading to incomplete data (Reps et al. 2021)

Table 2 GRADE components for raising scores

Components	Description
Large effect	When the Relative Risk (RR) is greater (or smaller) than a certain threshold, with no plausible confounders, the effect is considered large. When it is *much* greater (or smaller) than this threshold, combined with low risk of bias and no precision issues, it indicates a very large effect (Guyatt et al. 2011i). Researchers may also consider how quickly the effect appears compared to the typical course of the condition (Guyatt et al. 2011i).
Dose-response	The presence of a dose-response gradient is a criterion to believe there is a cause-effect relationship (Guyatt et al. 2011i).
All plausible residual confounding would reduce the effect	When all confounders and biases possibly affecting the study would lead to an underestimation of the effect (Guyatt et al. 2011i). Also, when the study found no effects, and all plausible confounders would suggest a spurious effect (Guyatt et al. 2011i).

- **Low:** We have limited confidence in the effect estimate, as it may differ substantially from the true effect.
- **Very low:** We have very little confidence in the effect estimate, as it will likely differ substantially from the true effect.

2.1.3 Evidence Versus Recommendations

GRADE distinguishes between two perspectives that require assessment. One reflects the role of a systematic review author who evaluates *the strength of evidence*, aiming to determine how confident one can be that the effect estimates of interventions are accurate. The other reflects the role of someone who formulates and evaluates *recommendations*

for practitioners, aiming to determine how confidently the effect estimates can support a specific recommendation (Balshem et al. 2011).

In the latter case, reviewers must consider the trade-offs between desirable and undesirable consequences[3] of following the recommendation, along with its costs, risks, and the values of those affected (Balshem et al. 2011). Ultimately, this distinction means that a recommendation can be weak or discretionary even when the evidence is of high quality—for instance, when desirable and undesirable effects are finely balanced, or when individuals are likely to differ in their preferences across intervention options.

2.2 Understanding GRADE-CERQual

While GRADE applies to quantitative systematic literature reviews, GRADE-CERQual (Confidence in the Evidence from Reviews of Qualitative research) applies to qualitative systematic reviews, supporting their use in decision-making (Lewin et al. 2018b). Its first step involves defining the review findings through a qualitative synthesis of evidence from primary studies. These findings may describe a phenomenon or an aspect of it Lewin et al. (2018b), and can take the form of themes, categories, theories, or other analytical constructs (Lewin et al. 2018a). Next, we discuss how to create these review findings.

2.2.1 Summaries of Review Findings

A key step in assessing confidence using CERQual is summarizing the review findings, especially when the original findings are lengthy (Lewin et al. 2018a). Researchers must balance the level of detail and the usefulness of the findings. Lewin et al. (2018a) provide specific guidance for writing review findings. In summary, they recommend:

- (i) making review findings as explicit as possible, clearly describing hypothesized relationships and pathways between factors;
- (ii) adapting the level of detail to the nature of the finding and the style agreed upon by the review team; and
- (iii) ensuring that, when a review finding supports a recommendation, it includes information needed for that purpose—such as acceptability, feasibility, or other implementation considerations.

2.2.2 CERQual Components

GRADE-CERQual proposes evaluations based on four components that contribute to an overall assessment of the level of confidence that review findings faithfully represent a phenomenon or one of its aspects (Lewin et al. 2018a). Table 3 briefly describes each one of them.

Researchers assess each component separately for every review finding, not for the synthesis as a whole (Lewin et al. 2018a). They begin by assigning high confidence to the finding and downgrade it when they identify concerns in one or more components. The levels of concern are:

[3] In terms of the outcomes of interest (Andrews et al. 2013).

Table 3 CERQual components for lowering scores

Components	Description
Methodological limitations	Degree of flaws in the design and execution of the primary studies that support the review finding (Munthe-Kaas et al. 2018). Researchers must select a quality appraisal tool, such as CASP (Critical Appraisal Skills Programme) (Dybå and Dingsøyr 2008a), or an evidence stardand such as the ACM SIGSOFT Empirical Standards for Software Engineering*.
Coherence	Extent to which the fit between the data from the primary studies and the review finding is clear and cogent (Colvin et al. 2018).
Adequacy of data	How rich and how much data support the review finding (Glenton et al. 2018)
Relevance	Extent that the body of evidence for the review finding applies to the research context, as specified in the research question (Noyes et al. 2018).

*https://www2.sigsoft.org/EmpiricalStandards/

- No or very minor concerns, which are unlikely to reduce confidence in the review finding.
- Minor concerns, which may reduce confidence in the review finding.
- Moderate concerns, which will probably reduce confidence in the review finding.
- Serious concerns, which are very likely to reduce confidence in the review finding.

For example, if a finding is supported by only one study and very few informants, reviewers would likely have serious concerns about data adequacy. Assuming no concerns in the other components, they would downgrade the finding from high to moderate or low confidence. Furthermore, reviewers must describe all relevant concerns in detail for each component, except minor ones (Lewin et al. 2018b). The assessment results typically appear in EP (Evidence Profile) and SoQF (Summary of Qualitative Findings) tables.

2.3 Related Work

As we discussed in Section 1, Dybå and Dingsøyr (2008a) were the first to delve into strength assessment of evidence in SE, elucidating the 2004 version of GRADE. That same year, the authors conducted a qualitative systematic literature review on agile software development, using GRADE to assess the strength of evidence (Dybå and Dingsøyr 2008b)—a study we included in our tertiary review. These works served as the field's initial references on the topic. However, early applications did not fully align with GRADE's original principles. For example, researchers provided generic strength evaluations instead of assessing each review finding individually (Kitchenham et al. 2022) or each "critical outcome of interest," as the method requires (Schünemann et al. 2023).

Kitchenham et al. (2015) emphasized that GRADE focuses on assessments of evidence strength in systematic literature reviews of randomized controlled trials (equivalent to controlled field experiments), a design that is difficult to implement in SE. They proposed using GRADE but considering SE's diversity of study types as the basis for assigning strength to findings. In their classification: (i) feasibility studies yield very weak evidence, even when methodologically sound; (ii) post-hoc analyses of industry datasets produce slightly stronger, yet still weak, evidence; (iii) laboratory experiments and quasi-experiments provide moderate strength evidence; and (iv) randomized field experiments offer the strongest

kind of evidence. This classification fits SE better but lacks detailed guidance on how to apply GRADE to each study type. For instance, how should researchers apply different downgrading or upgrading components depending on the study type, as GRADE allows?

Kitchenham et al. (2022) proposed the SEGRESS checklist for reporting secondary and tertiary studies in SE. It includes items about the strength assessment of evidence, a brief description of GRADE and GRADE-CERQual, along with examples of strength assessments of evidence in their supplementary material. Their explanations and examples for qualitative findings emphasize that the assessments should be at the *individual level* of review findings in line with the recommendations of GRADE and GRADE-CERQual. Previous assessments of confidence in SE led to a single evaluation of all findings taken together. However, Kitchenham et al. (2022) do not address all the details to enable effective strength assessments, such as the granularity of review findings (as discussed in Section 2.1.1), which is an important step when applying GRADE-CERQual. They also did not examine the current state of secondary studies regarding strength assessments.

Our tertiary review advances previous work by analyzing how SE secondary studies assess the strength of evidence in their findings. We present the current state of the field, reporting which methods researchers use, how they adapt them, and whether they derive recommendations to practitioners based on these assessments. Thus, we examine the extent to which SE researchers follow the recommendations of Dybå and Dingsøyr (2008a) and Kitchenham et al. (2015), or the examples in Kitchenham et al. (2022). In the next section, we present the methodology we used to conduct our study.

3 Research Methodology

This section presents the research methodology we followed for this tertiary review. We proposed the following research questions to understand the current strength assessment of review findings of systematic literature reviews and systematic mapping studies (SMS) in SE.

RQ 1 - Which methods are used for **grading** the strength of evidence in Software Engineering?

- RQ 1.1 - What are the **methods** for assessing the strength of evidence and the key papers describing them?
- RQ 1.2 - Does the method match the **review type**?

Answering RQ 1.1 involves mapping the methods researchers most commonly use and the references that they rely on for strength assessment. This analysis helps determine whether researchers are using the most recent versions of GRADE. RQ 1.2 complements this mapping by clarifying whether researchers apply the methods based on the review type. For example, we expect more recent qualitative systematic literature reviews to employ GRADE-CERQual, which became available in 2018. Beyond identifying which methods SE researchers use, we also aim to investigate whether and how they adapt them:

RQ 2 - How are researchers **using and/or adapting** the existing strength of evidence assessment methods to the Software Engineering context?

- RQ 2.1 - How are review findings **summarized**?
- RQ 2.2 - How are review findings **assessed**?

RQ 2.1 addresses the critical step of summarizing review findings in qualitative systematic literature reviews or identifying outcomes of interest in quantitative ones. Without clear summaries or outcome definitions, researchers cannot meaningfully assess the strength of evidence for each finding, as the approaches in Section 2 recommend. RQ 2.2 examines how SE researchers conduct these assessments in practice. Do they evaluate review findings individually? Which assessment components do they use?

As GRADE is also about recommendations, we also included a research question to address them in the context of SE:

RQ 3 - How do researchers draw **explicit recommendations** from evidence in the Software Engineering context?

- RQ 3.1 - Are recommendations directly **derived** from the systematic literature reviews' findings?
- RQ 3.2 - What are the **recommendation types** made due to the systematic literature reviews' findings (unhelpful, helpful with no need for grading, need grading)?
- RQ 3.3 - Are recommendations presented with a **strength assessment**? If so, how is the strength assessment carried out (balance between desirable and undesirable consequences, stakeholders' values and preferences, resource use)?

RQ 3.1 examines whether SE systematic literature reviews translate evidence into meaningful recommendations for practitioners. RQ 3.2 asks whether those recommendations are actionable and specific enough for practitioners to apply the evidence in practice. RQ 3.3 investigates whether researchers assess the recommendations, as GRADE proposes.

In summary, our goal is to examine how strength assessments of evidence are currently conducted in SE. To achieve this, we identified SE secondary studies that claimed to perform such assessments (as detailed in the next subsections) and analyzed whether they followed existing approaches strictly or introduced adaptations. When adaptations were present, we included illustrative examples to showcase how these approaches were modified (in Section 4). To satisfy this goal and answer to our research questions, we executed a tertiary study, whose search strategy we report next.

3.1 Search Strategy

Our search strategy mixes automated search and snowballing. For the automatic search, we first ran a pilot search on Scopus using the string "strength of evidence", restricting to the "computer science" area. Table 4 presents the set of four papers that we knew assessed the strength of evidence of their findings and that we used to evaluate our pilot search.

Table 4 List of known set of papers to assess the search string

Paper ID	Reference	Source
S1	Ali et al. (2010)	Previous study Budgen et al. (2020)
S2	Dybå and Dingsøyr (2008b)	Previous knowledge (reference paper on the topic)
S4	Matsubara et al. (2023)	Previous knowledge (a paper of ours)
S8	Selleri Silva et al. (2015)	Previous study Budgen et al. (2020)

The paper ID corresponds to the final IDs of papers after we finished all the selection process (see Table 9 for the complete list)

Table 5 String for the Automated Search

"strength of evidence" OR "strength of the evidence" OR "evidence strength" OR "strength assessment" OR "assessment of strength" OR "assessment of the strength" OR cerqual

We found three of these papers in this pilot search and decided to improve the search string to find all papers in our known set. Therefore, we changed to the string in Table 5.

Figure 1 shows the PRISMA flow diagram[4] with the details for our search and selection. After the pilot, we searched Scopus, ACM, IEEE, and Compendex for papers explicitly stating they assessed the evidence's strength ("Records identified from databases" in Fig. 1). We executed the automated search using title, abstract, and keywords. We restricted the search in Scopus to the "computer science" area to increase the precision of results. Allowing results from any area would lead to many spurious papers.

Our search strategy also involved a forward snowballing of the paper by Dybå and Dingsøyr (2008a) ("Records identified from snowballing" in Fig. 1). The rationale is that SE researchers will likely cite this paper to justify a strength assessment of the findings of their systematic literature reviews since it was the first paper in SE about the topic. We also considered six additional papers (also "Records identified from snowballing" in Fig. 1) that performed strength assessment of evidence, according to Trinkenreich et al. (2022). We had a total of 1310 papers as a result. Next, we eliminated the duplicates and the ones not accessible in full-text, getting to a total of 1116 papers for selection, which the next section details.

3.2 Selection Strategy

The selection strategy involved three filters, based on the inclusion and exclusion criteria in Table 6. We included any secondary study that claimed to assess the strength of evidence. For this reason, we also selected systematic mapping studies, even though strength assessments are not originally intended for this type of review. To be selected, a paper had to meet both inclusion criteria and violate none of the exclusion criteria. We defined EC01 and EC02 as the exact negations of IC01 and IC02 to clearly track the criteria used to include or exclude papers in each filter.

In the first filter ("records screened" in Fig. 1), we read titles and abstracts of all retrieved papers. We excluded any paper that met any exclusion criterion, except EC02, which we applied only in the second and third filters, and included any paper that satisfied IC01 (i.e., it reports a SE secondary study) or for which we had doubts. In other words, we chose to be highly inclusive at this stage, considering that a paper might assess the strength of evidence without stating so explicitly in its title, keywords, or abstract. This first filter yielded 94 papers.

[4] https://www.prisma-statement.org/prisma-2020-flow-diagram

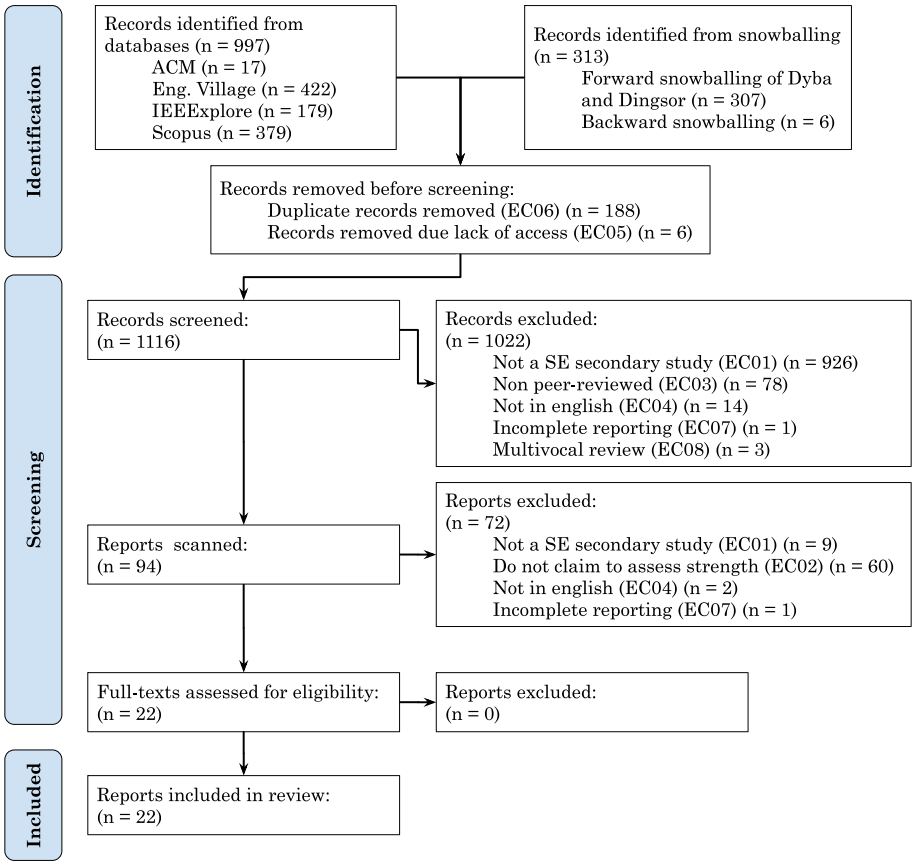


Fig. 1 PRISMA flow diagram showing the details of the search and selection

Table 6 Inclusion (IC) and exclusion (EC) criteria

ID	Criteria	Filter
IC01	The paper is a Software Engineering secondary study (systematic literature mapping/review), reporting at least the search, selection and extraction strategies, and the results.	1
IC02	The paper claims to have made an assessment of the strength of evidence supporting their findings.	2, 3
EC01	The paper is not a Software Engineering secondary study (systematic literature mapping/review) or does not report at least the search, selection, and extraction strategies.	1, 2
EC02	The paper does not assess the strength of evidence of review findings. For instance, it might rely only on quality assessments of individual primary studies, instead of assessing findings supported by the primary studies.	2, 3
EC03	The paper presents non-peer-reviewed results, e.g., in a book or grey literature.	1
EC04	The paper is not written in English.	1, 2
EC05	The paper is not accessible in full-text online.	1
EC06	The paper is a duplicate or a previous version of another already selected paper.	1
EC07	The systematic literature review or SMS is incomplete and not finished (only a protocol, for instance).	1, 2
EC08	The paper presents a multivocal literature review.	1, 2, 3

In the second filter, we scanned the full text of all papers included in the first filter ("records scanned" in Fig. 1). Specifically, we searched the texts for the terms "strength", "evidence", "low", "moderate", "high",[5] "grade", and "cerqual". When we found these terms, we examined their surrounding context. We also searched for explicit citations of Dybå and Dingsøyr (2008a) and for sections that presented or discussed strength assessments of findings. In short, we looked for evidence that justified including a paper under IC02—or, conversely, for signs that it met exclusion criterion EC02.

To increase the reliability of this step, both authors independently selected a subset of the papers in the second filter. To do so, the first author performed the selection of the papers alone. Next, she randomly selected a set of ten papers for kappa assessment, making sure this subset had both included and excluded papers (considering her selection: four included and six excluded). Next, the second author analyzed each of these papers separately, with no knowledge of the first author's decisions. Following, the first author collected the second author's decision. We agreed on all the papers to include in the subset. Because of this, we reached a kappa of 1, indicating very good/almost perfect agreement (Kitchenham et al. 2015). This process resulted in 22 papers meeting our criteria at this stage.

In the third filter ("full-texts assessed for eligibility" in Fig. 1), we read the full text of all remaining papers so far. If we found that a paper no longer satisfied the inclusion criteria, we excluded it. However, no papers were excluded at this stage, so we retained the same 22 papers from the previous filters. Table 9 (in Section 4) lists the 22 papers selected for data extraction and analysis, which we describe in the next section.

3.3 Extraction and Data Analysis Strategy

We extracted the data that Table 7 shows to answer the research questions.

As we extracted data from the primary studies using the extraction form, we also classified papers considering the categories of Table 8.

One author performed the initial data extraction and classification based on Table 8. The second researcher then reviewed the extracted data from two papers to validate the process and verify the classifications. We also applied a test-retest procedure to one paper, repeating the extraction four months after the original. The next section presents the results of this classification.

4 Results and Illustrative Examples

Table 9 presents the list of selected papers according to the procedures we reported in Section 3.3. Each paper has a unique identifier of the type S#, where # ranges from 1 to 22.

The remainder of this section answers the research questions from Section 2.1.1. Section 4.1 shows which methods SE researchers use to assess the strength of evidence for their review findings (RQ 1.1) and discusses whether such methods match the review type (RQ 1.2). Section 4.2 reports how SE researchers summarize review findings from SE systematic literature reviews and systematic mapping studies (RQ 2.1), and which

[5] The adjectives "low", "moderate", "high" are usually used to describe the strength of evidence.

Table 7 Data extracted from the papers

Data	Description	RQ
Review type	The review type can be Systematic Literature Review (quantitative, qualitative) or Systematic Mapping Study	RQ 1
Strength assessment method	Method used for strength assessment. It can be informal, GRADE, or GRADE-Cerqual, or any new method from another research area.	RQ 1
Outcome of interest [Quantitative systematic literature reviews]	For reviews addressing intervention effects, identify the critical outcomes. These may include specific quality or productivity measures. Quantitative systematic literature reviews are expected to define outcomes of interest.	RQ 2
The phenomenon of interest [Qualitative systematic literature reviews]	The phenomenon the study aims to describe or explain. Especially relevant in qualitative systematic literature reviews.	RQ 2
Review finding summarization [Qualitative systematic literature reviews]	Describe and exemplify how researchers summarized review findings. Are they clearly stated? Do they make relationships among variables explicit? What is the level of detail? This is especially relevant for qualitative systematic literature reviews.	RQ 2
Table to summarize review findings?	Does the paper include a table summarizing the review findings? Answer: yes/no, and the Table number in the paper, if yes.	RQ 2
Assessment dimensions	Dimensions used for the strength assessment. Report whether authors used GRADE or GRADE-CERQual as is, or adapted them. Also, note any custom dimensions used in informal assessments.	RQ 2
Recommendations from evidence	Recommendations derived from the review findings, if any. Also, the recommendation type and strength (against or in favor of an intervention). The type of recommendation can be: unhelpful, helpful with no need for grading, need grading.	RQ 3
Factors for strength assessment of recommendations	Criteria considered when assessing recommendation strength, e.g., analysis of trade-offs between benefits and harms, stakeholder values, or required resources.	RQ 3

Table 8 Classification scheme for the papers

Year	**Venue Type**	**Review Type**	**Method**	**RF summarized**	**Associates studies**
Publ. year	Journal or conference	SMS, quali systematic literature reviews or quanti systematic literature reviews	GRADE, GRADE-CERQual, other	Explicit, implicit, characterization of primary studies	Yes, for some RFs, No
RF table	**RF assessment**	**Dimensions**	**Recomm. from evidence**	**Recomm. type**	**Recomm. strength**
Yes, no	Generic, Specific	Method and version, or adapted dimensions	Specific, Generic, or Not explicit	Unhelpful, helpful that do not require grading, or helpful that require grading	Yes, no

RF = Review Finding

dimensions the researchers used in strength assessment (RQ 2.2). Section 4.3 reports how SE researchers derive recommendations from their review findings (RQ 3.1), the types of recommendations the researchers made (RQ 3.2), and whether they also assess the strength of such recommendations (RQ 3.3).

Table 9 List of selected papers and their classification

ID	Paper title	Reference
S1	A systematic review of comparative evidence of aspect-oriented programming	Ali et al. (2010)
S2	Empirical studies of agile software development: A systematic review	Dybå and Dingsøyr (2008b)
S3	Global Software Engineering: Evolution and Trends	Ebert et al. (2016)
S4	Much more than a prediction: Expert-based software effort estimation as a behavioral act	Matsubara et al. (2023)
S5	On strategies for testing software product lines: A systematic literature review	Machado et al. (2014)
S6	Requirements engineering for software product lines: A systematic literature review	Alves et al. (2010)
S7	Requirements modeling languages for software product lines: A systematic literature review	Sepúlveda et al. (2016)
S8	Using CMMI together with agile software development: A systematic review	Selleri Silva et al. (2015)
S9	Women's participation in open source software: A survey of the literature	Trinkenreich et al. (2022)
S10	A systematic review on the engineering of software for ubiquitous systems	Sánchez Guinea et al. (2016)
S11	Model-based testing for software safety: a systematic mapping study	Gurbuz and Tekinerdogan (2018)
S12	State of the art of agile governance	de O.Luna et al. (2014)
S13	25 Years of Requirements Engineering in Brazil: A Systematic Mapping	Oliveira et al. (2013)
S14	A systematic review of evaluation of variability management approaches in software product lines	Chen and Babar (2011)
S15	Critical Issues on Test-Driven Development	Kollanus (2011)
S16	Investigating the factors which impact SPI implementation initiatives in software SMEs—A systematic map and review	Sharma and Sangal (2019)
S17	Open innovation in software engineering: a systematic mapping study	Munir et al. (2016)
S18	Recovering from a decade: a systematic mapping of information retrieval approaches to software traceability	Borg et al. (2014)
S19	Spot pricing in the Cloud ecosystem: A comparative investigation	Li et al. (2016)
S20	Variability Modeling in Software Product Line: A Systematic Literature Review	Jaffari et al. (2021)
S21	A systematic mapping study on requirements scoping	Wnuk and Kollu (2016)
S22	Understanding and addressing quality attributes of microservices architecture: A Systematic literature review	Li et al. (2021)

4.1 RQ 1 - Which Methods are used for Grading the Strength of Evidence in Software Engineering?

Table 10 shows that SE researchers used varied methods for the strength assessment of review findings. Ten papers (45.6% of the papers that we included in our study) employed GRADE for strength assessment. In all these papers, the researchers referenced the 2004 version (Atkins et al. 2004), even in cases of secondary studies on SE published after the release of papers reporting a more recent version of the approach. Additionally,

Table 10 Papers per review type and strength assessment method

	Method				
Review Type	GRADE	GRADE-CERQual	Hierarchy of evidence	Primary studies' quality	None
SMS	S6-7, S10-12		S5	S13	S18, S20
Quanti systematic literature review	**S1**				
Quali systematic literature review	S2, S8-9	**S4**	S3	S16, S17, S19, S21, S22	
Mixed	S14				S15

Papers in boldface were the ones using an appropriate method given the review type

one paper using GRADE only mentioned one of its components, emphasizing that the results of the primary studies were consistent. Therefore, it did not employ the method fully.

Additionally, one paper (4.5%) applied GRADE-CERQual, two (9.0%) used a hierarchy of evidence, and six (27.3%) relied on the quality of primary studies—corresponding to only one component of strength assessment in GRADE or GRADE-CERQual. Three papers (13.6%) did not apply any specific method. One of them (S15) noted that the issues with their evidence resembled those reported by Dybå and Dingsøyr (2008b), which led them to classify the evidence strength as very low. The other two assessed elements the researchers considered relevant, such as dataset characteristics and evaluation context (S18), or types of assessment (e.g., illustrative examples, academic studies, or industrial practice; S20).

> **RQ 1.1** - What are the **methods** for assessing the strength of evidence and the key papers describing them?
> **Answer:** The 2004 GRADE version is the most referenced method by SE researchers for strength of evidence assessments. However, many studies also base such assessments only on the quality of primary studies, which is only one of the components of more sophisticated assessment methods.

Table 10 also shows that nine (40.9%) of the secondary studies in our tertiary review are SMSs. These studies categorize primary studies into high-level categories/themes relevant to the research topic, often reporting the percentage of studies within each category. Ten studies (45.6%) were qualitative systematic literature reviews, meaning their findings aim to characterize a phenomenon of interest. Only one paper (4.5%) was a quantitative systematic review, with clearly defined outcomes of interest and a synthesis of evidence related to those outcomes. Two papers (9.0%) were mixed reviews, combining features of both qualitative and quantitative approaches.

Based on this classification, only two papers (9.0%) used a strength assessment method consistent with their review type, as Table 10 shows. S1 (Ali et al. 2010) is a quantitative systematic literature review in which the authors applied GRADE, and S4 (Matsubara et al. 2023) is a qualitative systematic literature review in which the authors used GRADE-CERQual.

RQ 1.2 - Does the method match the **review type**?
Answer: Only two of 22 papers chose a method that matched their review type. Also, nine SMSs claimed to make strength assessments of evidence, though they are not meant for this review type.

4.2 RQ 2 - How are Researchers using and/or Adapting the Existing Strength of Evidence Assessment to the Software Engineering Context?

This section presents issues related to the format and presentation of review findings. We analyzed how researchers summarized them. We also examined whether they established a clear **association** between each review finding and its supporting primary studies, and whether they included a **table or figure** to summarize the findings. Moreover, this section describes how authors assessed the strength of their review findings. We considered whether the assessment was **specific** to each review finding, **generic** across all findings, or limited to evaluating the quality of the primary studies only. Additionally, instead of describing each study in detail, we provide illustrative examples to show how researchers reported their assessments.

4.2.1 RQ 2.1 - How are Eeview Findings Summarized?

Table 11 shows whether and how papers summarized their review findings—through **explicit** summaries, **implicit** descriptions in the text, or by **characterizing primary studies**. It also indicates whether the findings appeared in **tables/figures** or in **textual format** only, and whether authors clearly **linked findings to their supporting primary studies**. In the remainder of this section, we illustrate these patterns using examples from selected papers based on the characteristics that Table 11 presents.

Twelve studies (54.5%) present **explicit** review findings. S4, a qualitative systematic literature review, illustrates this desirable practice. It reports its findings in **tables** and clearly **associates** each finding with its supporting studies. Figure 2 shows one of its review findings, highlighting all these characteristics.

S3 is another example of a qualitative systematic literature review with **explicit** review findings presented in a **table**. Unlike S4, however, the authors do not indicate the specific primary studies **associated** with each finding. We did not include a figure illustrating its findings because we could not obtain permission to reproduce or adapt the original.

Only one study (4.55%), S8, presented **implicit** review findings, as Fig. 3 shows. From the research question and the results section, one can infer that combining CMMI

Table 11 Papers considering the summarization and association between review findings and supporting studies

	Table or figure		Textual-only	
	With studies	**No studies**	**With studies**	**No studies**
Explicit review finding	S1, S4, S9, S14-S17, S19, S21, S22	S3	None	S2
Implicit review finding	None	None	None	S8
Characterization of primary studies	S6, S10, S12	S13, S18	S5, S11	S7, S20

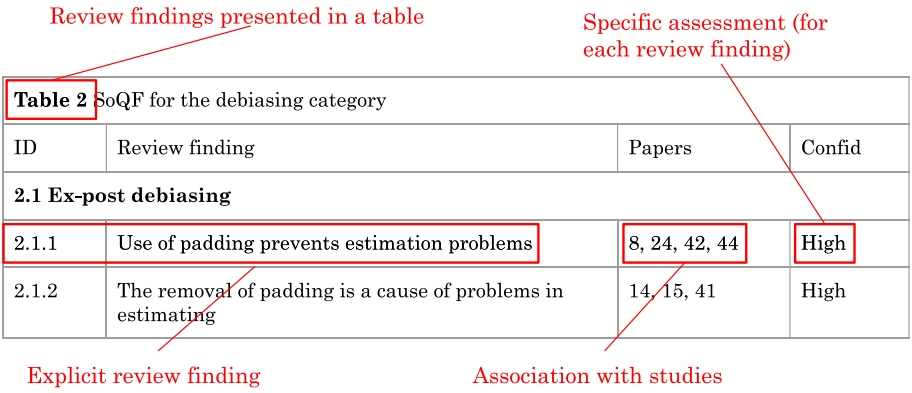
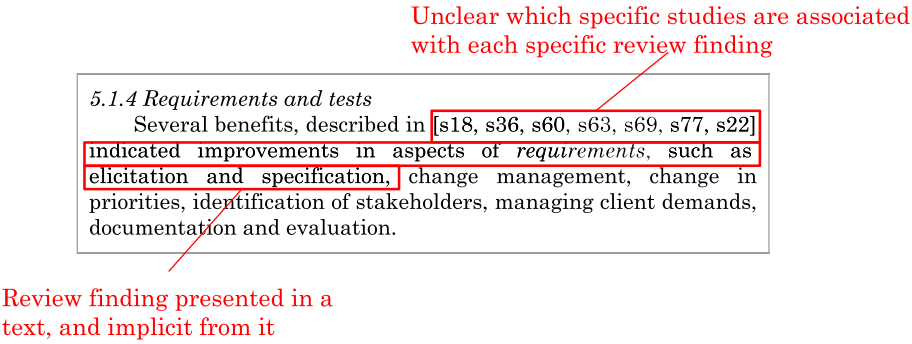


Table 2 SoQF for the debiasing category

ID	Review finding	Papers	Confid
2.1 Ex-post debiasing			
2.1.1	Use of padding prevents estimation problems	8, 24, 42, 44	High
2.1.2	The removal of padding is a cause of problems in estimating	14, 15, 41	High

Fig. 2 Example of review finding from S4. The original table in Matsubara et al. (2023)—"Reprinted from Empirical Software Engineering, 28, Patrícia G. F. Matsubara, Igor Steinmacher, Bruno Gadelha, Tayana Conte. Much more than a prediction: Expert-based software effort estimation as a behavioral act, Article n°98, 48 pages, Copyright (2023), with permission from Springer Nature"



5.1.4 Requirements and tests

Several benefits, described in [s18, s36, s60, s63, s69, s77, s22] indicated improvements in aspects of *requirements*, such as elicitation and specification, change management, change in priorities, identification of stakeholders, managing client demands, documentation and evaluation.

Fig. 3 Example of review finding from S8. Original text from Selleri Silva et al. (2015)—"Reprinted from Information and Software Technology, 58, Fernando Selleri Silva, Felipe Santana Furtado Soares, Angela Lima Peres, Ivanildo Monteiro de Azevedo, Ana Paula L.F. Vasconcelos, Fernando Kenji Kamei, Silvio Romero de Lemos Meira, Using CMMI together with agile software development: A systematic review, 20-43, Copyright (2015), with permission from Elsevier"

and agile software development improves requirements elicitation and specification. However, S8 does not clearly **associate** each finding with the supporting primary studies.

S1 was the only quantitative systematic literature review among the included papers. It compares Aspect-Oriented Programming (AOP) with non-AOP approaches, focusing on outcomes such as performance and code size. Figure 4 shows an excerpt from the **table** presenting its findings, which are **explicitly** derived from the research questions and outcomes of interest. The authors clearly **associated** the observed effects of AOP on each outcome with the corresponding primary studies.

Nine papers (40.9%) in our review were SMSs, such as S10. These studies do not include review findings that characterize a phenomenon or evaluate the effects of an intervention on defined outcomes. Instead, SMSs report findings based on the **characterization of primary**

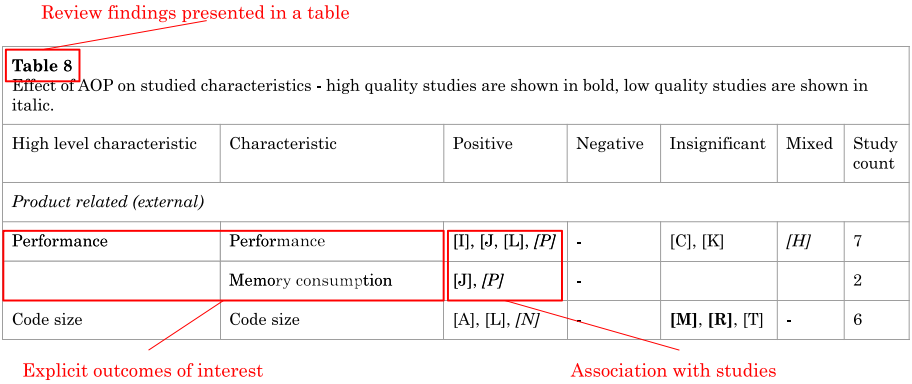


Table 8
Effect of AOP on studied characteristics - high quality studies are shown in bold, low quality studies are shown in italic.

High level characteristic	Characteristic	Positive	Negative	Insignificant	Mixed	Study count
Product related (external)						
Performance	Performance	[I], [J, [L], *[P]*	-	[C], [K]	*[H]*	7
	Memory consumption	[J], *[P]*	-			2
Code size	Code size	[A], [L], *[N]*	-	**[M]**, **[R]**, [T]	-	6

Fig. 4 Example of review finding from S1. The original table in Ali et al. (2010)—"Reprinted from Information and Software Technology, 52/9, Muhammad Sarmad Ali, Muhammad Ali Babar, Lianping Chen, Klaas-Jan Stol. A systematic review of comparative evidence of aspect-oriented programming, 871-887, Copyright (2010), with permission from Elsevier"

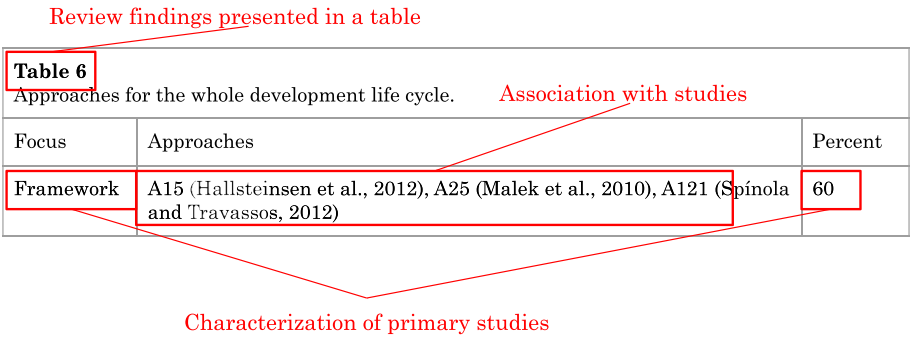


Table 6
Approaches for the whole development life cycle.

Focus	Approaches	Percent
Framework	A15 (Hallsteinsen et al., 2012), A25 (Malek et al., 2010), A121 (Spínola and Travassos, 2012)	60

Fig. 5 Example of review finding from S10. The original table is in Sánchez Guinea et al. (2016)—"Reprinted from The Journal of Systems and Software, 118, Alejandro Sánchez Guinea, Grégory Nain, Yves Le Traon, A systematic review on the engineering of software for ubiquitous systems, 251-276, Copyright (2016), with permission from Elsevier"

studies. Figure 5 shows part of a **table** from S10 that categorizes the studies, listing each **associated** paper and the percentage of papers within each category.

Another SMS included in our review was S7. Like S10, it classifies papers into categories and reports the percentage of primary studies in each. However, as Fig. 6 illustrates, it does not list the specific studies per category—i.e., papers are not **associated** with the categories.

RQ 2.1 - How are **review findings** summarized?
Answer: Most papers (12) present explicit summaries of their review findings, which helps readers clearly identify the results. Another nine papers characterize primary studies, as they are systematic mapping studies. A majority (16) summarize their findings using tables or figures rather than relying solely on text. Additionally, most papers (15) clearly associate each finding with the supporting studies—an essential practice for clarifying the evidence base and enabling strength assessment.

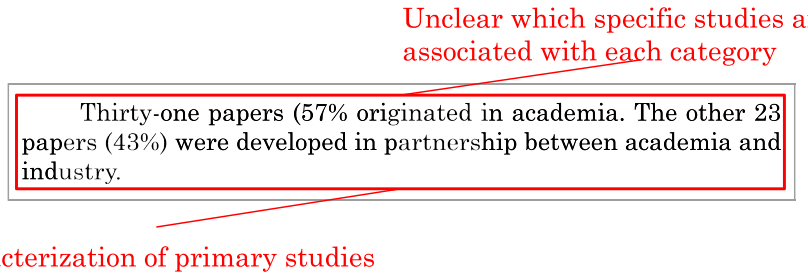


Fig. 6 Example of review finding from S7. The original text is from Sepúlveda et al. (2016)—"Reprinted from Information and Software Technology, 69, Samuel Sepúlveda, Ania Cravero, Cristina Cachero, Requirements modeling languages for software product lines: A systematic literature review, 16-36, Copyright (2016), with permission from Elsevier"

4.2.2 RQ 2.2 - How are Review Findings Assessed?

This section examines how SE researchers conducted and reported their strength assessments. Specifically, we focus on the level of assessment and the components used. Twenty papers (90.9%: S1, S2, S5–S22) applied a **generic** evaluation for all review findings, which diverges from GRADE and GRADE-CERQual guidelines. Only two papers (9.0%: S3 and S4) followed the recommended approach by providing specific assessments for each review finding. In the remainder of this section, we explore these aspects in more detail using examples.

Generic Versus Specific Assessments S9 is a qualitative review that applied GRADE and conducted a **generic** assessment, evaluating the set of findings **as a whole** using the method's dimensions. Figure 7 shows one of its review findings. Although the finding is presented individually, the authors did not provide a specific strength assessment for it—even though the format would have supported such an evaluation.

The **generic** assessment of S9 appears in a dedicated section of the paper—a common pattern among studies that follow this approach. To illustrate, the authors addressed the GRADE dimension of consistency with the following statement:

> "*Consistency (...) Since the outcomes of primary studies were not presented in a comparable way and reporting protocols vary according to the study, evaluating the synthesis of quantitative results could be misleading. However, having several studies grouped into each strategy (see Fig. 11 reflects positively on the consistency of the studies and we conclude that, in general, strategies are consistent*" (Trinkenreich et al. 2022).

S3 and S4 were the only papers that assessed the strength of individual review findings—a practice we refer to in this study as a **specific** assessment. We use S4 to illustrate this level of assessment. The authors of S4 applied GRADE-CERQual without deviating from its proposed components (Section 2.2). In other words, they assessed each review finding

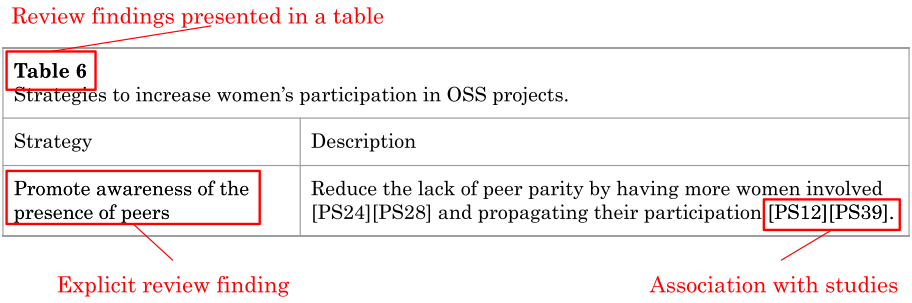


Fig. 7 Example of review finding from S9. The original table is in Trinkenreich et al. (2022). Reprinted from ACM Transactions on Software Engineering and Methodology, 31, 4, Article n°81, 37 pages, Copyright (2022), with the authors' permission

individually, considering the **methodological limitations** of the supporting primary studies, the **coherence** of the data they provided, the **adequacy** of that data, and the **relevance** of the study expected contexts from the research questions compared to the real-world settings of the phenomenon under investigation. Although Fig. 2 does not display these assessment components, the authors included them in their supplementary material (Fig. 8).

There were four studies supporting the review finding, which concerns padding, i.e., the addition of a contingency buffer to estimates to handle unexpected events (Matsubara et al. 2023). Figure 8 shows the specific data used to construct this finding in the first column. The reviewers reported no methodological limitations, using an instrument adapted from Dybå and Dingsøyr (2008a) to assess the quality of the supporting primary studies. They also identified no coherence issues: all studies addressed padding directly, although one did not examine the relationship between padding and problem prevention. Since all studies provided evidence *in favor* of padding, concerns about coherence were further reduced. The reviewers also reported no concerns about data adequacy, as the studies offered rich data about the phenomenon. They found no relevance concerns either, given the diversity of SE contexts, the inclusion of recent studies, and the involvement of software practitioners as research participants.

Strength Assessment Components Another detail of how the assessment was carried out relates to the components of the strength assessment. GRADE has five components for downgrading evidence: risk of bias, publication bias, precision, directness, and consistency. Nine (40.9%) studies using GRADE in our review used a version that dates back to 2004 (Atkins et al. 2004). More importantly, the authors made an assessment similar to the one in Dybå and Dingsøyr (2008b), which involved four components only: study design, study quality, consistency, and directness. Such papers did not refer to the precision component nor to publication bias. Figure 9 shows how such components relate to the most recent GRADE version.

In addition, one paper (4.5%, S14) mentioned only a single GRADE component, offering an incomplete view of the evidence strength supporting its review findings: "(...) the reported evidence appears to be quite consistent, which is one of the four key elements that

Supporting papers

GRADE-CERQual components

Papers contributing to the review finding	Methodological limitations	Coherence	Adequacy	Relevance	CERQual assessment of confidence in the evidence
8, 24, 42, 44	No **methodological limitations**	**Minor concerns about coherence** All studies address padding directly. Study 44 do not address padding as preventing problems directly	**No concerns about adequacy** Paper 8: 5 companies, 4 estimation sessions, and 12 interviewees. Paper 24: 68 projects and 8 interviewees in one company. Paper 42: 62 participants. Paper 44: 112 participants	**No concerns about relevance** Wide range of software engineering contexts (industries and sizes). Participants are software engineering or information systems practitioners. Recent papers included (Ranging from 1991 to 2021)	High confidence

Fig. 8 Part of the Evidence Profile table of S4 (original table from the supplementary material in Matsubara et al. (2023))—"Reprinted from Empirical Software Engineering, 28, Patrícia G. F. Matsubara, Igor Steinmacher, Bruno Gadelha, Tayana Conte. Much more than a prediction: Expert-based software effort estimation as a behavioral act, Article n°98, 48 pages, Copyright (2023), with permission from Springer Nature"

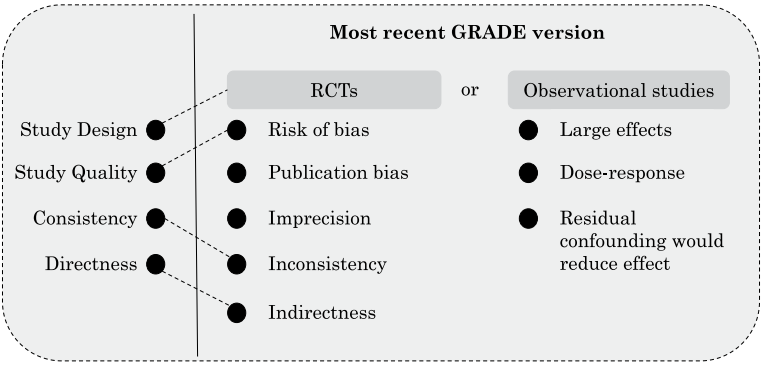


Fig. 9 The relationships between the components that Dybå and Dingsøyr (2008b) employed to the left of the dotted line and the most recent version of GRADE (2011) to the right

determine the strength of evidence according to the GRADE (Grading of Recommendations Assessment, Development and Evaluation) system (...)" (Chen and Babar 2011).

As Table 10 shows, only one paper (4.5%, S4) used GRADE-CERQual. The authors applied all original components of the method in their strength assessment. In addition, two papers (9.0%, S3 and S5) used a hierarchy of evidence. This approach does not rely on explicit dimensions but rather categorizes studies based on their design. For example, S3 applied this method by assigning the highest score to "multi-case, longitudinal or replication studies (can be lab only or with industry involvement)", which the authors regarded as providing strong evidence (Ebert et al. 2016). These scores contributed to the calculation of the average score for each theme.

Three papers (13.6%, S15, S18, and S20) used no established method to assess strength of evidence, despite the fact that two of them (S18 and S20) had specific research questions addressing this topic. The authors of S15 compared the limitations of their evidence with general concerns in SE, drawing on Dybå and Dingsøyr (2008b): "[Dybå and Dingsøyr (2008b)] concluded that the strength of empirical evidence on agile practices is very low. The conclusion is based on similar issues that are related to TDD research in this study." S18 evaluated strength based on: (i) dataset characteristics (e.g., size, origin, relevance); (ii) reporting quality; and (iii) evaluation context. S20 assessed strength by examining the type of evaluation in the primary studies–ranging from illustrative examples to academic or industrial practice (the latter considered the strongest form of evidence).

Another six papers (27.3%, S13, S16, S17, S19, S21, and S22) equated quality assessment of primary studies with strength assessment. Two of them (S16 and S17) had explicit research questions on evidence strength, and one (S19) included a section titled "evidence strength." Three of these papers (S16, S17, and S21) applied the quality instrument from Ivarsson and Gorschek (2011), which considers rigor and relevance explicitly, similar to GRADE-CERQual components of methodological limitations and relevance. However, this instrument is designed to assess individual primary studies—not bodies of evidence.

RQ 2.2 - How are **review findings** assessed?
Answer: Most papers (20) make a generic assessment for the whole set of review findings instead of a specific assessment for each review finding, thus deviating from the assessment guidelines. Moreover, most of the assessments using GRADE were inspired by Dybå and Dingsøyr (2008b), relying on four components: study design, study quality, consistency, and directness. However, GRADE proposes five dimensions for rating down evidence (risk of bias, publication bias, inconsistency, imprecision, and indirectness). Furthermore, many papers (six) confused the quality assessment of primary studies with strength assessment of bodies of evidence.

4.3 RQ 3 - How do Researchers Draw Explicit Recommendations from Evidence in the Software Engineering Context?

As we explained in Section 2.1.3, GRADE is also an approach for rating the strength of recommendations. Table 12 shows the mapping of SE secondary studies' characteristics and types of recommendations.

Most secondary studies (81.8%, 18) included in our review do not present clear and explicit recommendations based on the evidence gathered from primary studies. Nine of these are SMSs, which are not expected to provide recommendations. Among the remaining secondary studies, only two (9.0%, S4 and S9) clearly articulate specific recommendations derived from findings on the effects of an intervention. The quotation below illustrates a recommendation concerning the effect of padding as an intervention (see Fig. 2).

> "*Use padding for debiasing in large projects. Obs.: This can lead to overestimation, which is an undesirable outcome that decreases accuracy. Low resource usage and easy to implement. However, it can be hard to decide how much to pad to avoid overestimation*" (Matsubara et al. 2023).

Figure 7 provides another example of an explicit and specific recommendation derived from a review finding—-in this case, the recommendation coincides with the finding itself. Another study (4.5%, S3) also offers explicit and specific recommendations, although not directly based on findings about the effects of an intervention. This paper presents a matrix of risk mitigation strategies that were developed independently

Table 12 Distribution of papers per recommendation type and assessment

	Recommendation type			
Recommendations	Helpful (grading)	Helpful (no grading)	Unhelpful	NA
Explicit, specific, from review findings	S4, S9	S4, S9	S4	None
Explicit, specific, not from review findings	None	None	S3	None
Explicit, generic	None	None	S2	None
Not explicit	None	None	None	S1, S5-S8, S10-S22
Recommendations assessment	No			NA

from its review findings. One additional study (4.5%, S2) includes only generic recommendations in the "implications for research and practice" section:"*The evidence also suggests that agile methods not necessarily are the best choice for large projects. Thus, consistent with recommendations provided by others [11,12,15], we suggest that practitioners carefully study their projects' characteristics and compare them with the relevant agile methods' required characteristics*" (Dybå and Dingsøyr 2008b).

RQ 3.1 - Are recommendations directly **derived** from the systematic literature reviews' findings?
Answer: Many papers do not present any recommendations. In only two, the authors clearly derived them from the synthesized review findings. In another, the recommendations were drawn from primary studies but not from the synthesis itself. Yet another paper included only generic suggestions, embedded in the discussion of practical implications.

Table 12 presents our classification of recommendations by type, following GRADE terminology. According to Guyatt et al. (2011c), **unhelpful recommendations** are neither specific nor actionable, which may lead to inconsistent or counterproductive interpretations. **Helpful recommendations that do not require grading** are those for which the benefits clearly outweigh the undesirable effects—typically supported by substantial indirect evidence. In such cases, teasing out the supporting evidence would be difficult, and alternative options often appear implausible or even laughable (Guyatt et al. 2011c). In contrast, **recommendations that require grading** involve uncertainty about their net benefit, reasonable resource use, and other trade-offs.

The quotations from S2 and S4 discussed above exemplify unhelpful recommendations, as they lack specificity about project size thresholds for applicability. By contrast, the recommendation from S9 in Fig. 7 is more specific and actionable. We classified it as a helpful recommendation that does not require grading, as its benefits clearly outweigh any potential downsides. An example from S9 of a helpful recommendation that does require grading is "Create and enforce a Code of Conduct", further illustrated in the following quotation:

"*Develop a code of conduct [PS3] [PS16] [PS27] [PS47] as the collective norms on (un)acceptable behavior at all interactions, explicit the prohibition of harassment [PS21] [PS24] [PS37], and that violations have consequences [PS8] [PS33], while having mechanisms to invigilate the use and to apply the punishments accordingly if necessary.* Trinkenreich et al. (2022).

Such a recommendation might require grading the supporting evidence to determine whether it effectively increases women's participation in Open Source Software—the intended outcome of the intervention. As noted in S9, the authors highlight that implementing a code of conduct does not necessarily result in greater participation by

women compared to communities without one (Trinkenreich et al. 2022). Moreover, grading is also warranted given the potentially high resource demands associated with developing and enforcing such a code.

RQ 3.2 - What are the **recommendation types** made due to the systematic literature reviews' findings (unhelpful, helpful with no need for grading, need grading)?
Answer: Among the few papers that provide recommendations, there is a mix of types. Three offer unhelpful recommendations. Two present helpful recommendations that require no grading—and these same two also include helpful recommendations that do require grading.

Table 12 also shows that none of the papers providing recommendations assessed their strength. In other words, the authors did not evaluate key factors such as the balance between desirable and undesirable effects, stakeholders' values and preferences, or the resources required to implement the intervention. Such assessments should be grounded in studies to identify which preferences stakeholders may have regarding one intervention over another.

RQ 3.3 - Are recommendations presented with a **strength assessment**? If so, how is the strength assessment carried out (balance between desirable and undesirable consequences, stakeholders' values and preferences, resource use)?
Answer: No paper assessed the strength ofits recommendations, even when those recommendations were based on evidence.

5 Discussion

Assessing the strength of evidence enables systematic review authors to produce more credible and trustworthy results (Wohlin and Rainer 2021). In this section, we discuss the findings on evidence and recommendation strength assessments of Section 4.

5.1 Review types and Strength Assessment

SEGRESS, a recent guideline for reporting secondary and tertiary studies in SE, explicitly states that confidence assessments of the evidence body are essential for systematic reviews (Kitchenham et al. 2022). Still, few systematic reviews execute it. From a total of 13 systematic reviews (i.e.. excluding the nine systematic mapping studies) from our sample of selected papers, only six claim to use a structured approach to strength assessment (such as GRADE or GRADE-CERQual). Of those, only two applied a method adequate for their review type. As a reminder, while GRADE is appropriate for quantitative systematic reviews focused on outcomes of interest (Guyatt et al. 2011b; Kitchenham et al. 2022), GRADE-CERQual is the correct one for qualitative reviews (Lewin et al. 2018b; Kitchenham et al. 2022).

5.2 The Evolution of GRADE

Until now, the SE community has not discussed the impact of GRADE revisions on SE literature (Kitchenham et al. 2022). This work addresses this gap. Although the original version of GRADE was published in 2004—over twenty years ago—most papers in our review that used GRADE relied on this early version, despite subsequent updates in 2008 and 2011. While some delay in adopting updates is expected, the latest version offers more detailed guidance for systematic literature review producers (Guyatt et al. 2011a).

Furthermore, when applying GRADE "as is" (Section 5.8), many studies omitted key dimensions, such as precision (Section 4.2.2). This omission may reflect challenges in assessing certain components within the SE domain, which may also signal the difficulties of applying such a method in our field. One possible reason is the lack of standardized sets of outcomes of interest in our field, a point we explore further in Section 5.3. Additionally, many SE reviews assessed study design as a component, although GRADE uses it only to set the initial confidence level—high for randomized controlled trials and low for observational studies.

5.3 Synthesizing Evidence is an Issue of Concern

Individual primary studies provide "pieces of evidence", and EBSE aims to build a body of knowledge by aggregating these pieces to produce increasingly credible evidence on a given topic (Wohlin and Rainer 2021). Therefore, aggregation and synthesis are critical for SE research to effectively impact practice. A step toward better aggregation is the intelligent summarization of review findings. SE systematic literature reviews must generate explicit and specific findings to enable appropriate strength assessments.

To some extent, the challenge of synthesizing evidence relates to the concept of outcomes of interest. To synthesize quantitative systematic literature review findings about intervention effects, evidence from multiple primary studies must address the same outcomes—that is, the same dependent variables, or the same outcome measures (Kitchenham et al. (2002)). This alignment is necessary for statistical aggregation (e.g., meta-analysis) or even simpler methods like vote counting. In Software Engineering, this implies a need for greater consistency across primary studies in selecting and reporting outcomes for similar interventions. In medicine, the "core outcome set" initiative addresses this challenge by defining a minimum set of outcomes that all trials in a given area should report (Clarke and Williamson 2016), while still allowing researchers to report additional outcomes (Dodd et al. 2018). Similar initiatives may benefit the SE field.

Another issue is the presentation of findings. GRADE recommends the use of Summary of Findings (SoF) and Evidence Profile (EP) tables to present results per outcome of interest. GRADE-CERQual offers a parallel structure for qualitative review findings: Summary of Qualitative Findings and Evidence Profile tables. A Summary of Qualitative Findings table closely resembles the table in Fig. 2, and follows the structure in Table 13.

An Evidence Profile (EP) table expands on the Summary of Qualitative Findings table by including the CERQual dimensions—methodological limitations, coherence, adequacy, and relevance—as column headings (Lewin et al. 2018a)[Table 4]. In each of these columns, authors describe specific concerns about the corresponding component, if any, regarding the evidence supporting the review finding. Figure 8 provides an example.

Table 13 Headings of a Summary of Qualitative Findings table

Summary of RF	Studies contributing to the RF	CERQual assessment of confidence in the evidence	Explanation of CERQual assessment
Summary text. One line per RF.	List of the papers reporting the studies	High, moderate, low, or very low	Explanation of the assessment, including reasons for downgrading, based on CERQual dimensions.

RF = review finding. Adapted from Lewin et al. (2018a)

Many of the systematic literature reviews we reviewed present explicit findings. However, these are often organized as bullet-point lists (e.g., the findings in S3 about reasons for Global Software Engineering failures) or used as section headings (e.g., the themes in S2). When findings are implicit, as in S8, they become difficult to identify. Regardless of format, the field needs structured approaches for reporting review findings to facilitate their extraction and support evidence-based recommendations for practitioners (Budgen et al. 2020).

5.4 Quality Assessment of Primary Studies is not a Strength Assessment of Evidence

A few papers included in our review either posed research questions about strength assessment or mentioned it, but did not conduct an assessment of a body of evidence in the sense of GRADE or GRADE-CERQual. Upon closer inspection, we found that the authors of these reviews had instead focused on assessing the quality of the primary studies. Three of the papers that equated quality assessment with strength assessment used the instrument proposed by Ivarsson and Gorschek (2011), which includes two components: rigor and relevance. Two others adapted the checklist from Dybå and Dingsøyr (2008a) to assess quality.

It is important to clarify that while strength assessment following GRADE includes a component related to study quality, it goes beyond that. GRADE incorporates a "risk of bias" dimension, and GRADE-CERQual uses the "methodological limitations" dimension—both of which require evaluating the quality of the primary studies supporting a review finding. However, other dimensions in these approaches suggest that more is involved.

For example, GRADE-CERQual also calls for evaluating the coherence of findings, the adequacy of the data (i.e., its "quantity" and richness), and its relevance. While some SE quality instruments address relevance such as Ivarsson and Gorschek (2011), they more frequently emphasize characteristics related to rigor ("the scientificity, practicability and completeness of research method and result analysis"), rationality ("clear statement of research context and objective"), and credibility ("the validity, certainty and reliability of research findings and conclusions") (Yang et al. 2021). These characteristics align closely with the "methodological limitations" component but do not sufficiently address other dimensions of strength assessment.

Therefore, GRADE and GRADE-CERQual are comprehensive frameworks detailed in a large body of literature. While they may not explicitly name every issue relevant for evidence quality (e.g., p-hacking or HARKing), they do address several of them under well-defined and varied components. For instance, stopping early for benefit—often discussed in connection with p-hacking—and selective outcome reporting are explicitly considered under the GRADE "risk of bias" dimension. Likewise, the framework includes a dedicated component for publication bias, supported by its own methodological paper. Furthermore,

while GRADE does not use the term "construct validity," its indirectness and inconsistency domains capture mismatches between interventions, measures, and review questions—implicitly covering construct-related concerns, for instance.

Another key distinction between quality of primary studies and strength of evidence lies in the purpose of each assessment. Quality assessments of primary studies support review processes such as study selection or analysis (e.g., by excluding studies below a certain threshold). In contrast, strength assessments target the synthesized *body of evidence* supporting a review finding, as we discuss in the next section. In other words, strength assessments aim to judge the credibility of conclusions based on all the aggregated evidence from relevant primary studies.

5.5 Strength Assessments Should be Specific for each Review Finding

Furthermore, there is the issue of the unit of analysis. Kitchenham et al. (2022) discuss it, noting that quality assessments focus on individual primary studies, while strength assessments concentrate on the results of the synthesis or aggregation across multiple studies. Therefore, strength assessments should not be applied generically to the entire set of review findings, as many secondary studies in our review did. Each review finding requires its **own** strength assessment.

The rationale is that an intervention may influence a wide range of outcomes, leading to both desirable and undesirable effects. The amount and quality of supporting evidence can vary significantly across outcomes or review findings (Kitchenham et al. 2022). As a result, each outcome requires an individual strength assessment of the aggregated evidence regarding the intervention's effects, considering study designs and each GRADE dimension—a requirement for claiming proper use of GRADE (Schünemann et al. 2023). Likewise, in qualitative reviews, a phenomenon of interest may encompass multiple aspects described in separate review findings. The quality, quantity, richness, or relevance of supporting evidence can vary from one finding to another, justifying the need for individual assessments.

In some cases, the strength assessment was also incomplete. For example, Kollanus (2011) cited Dybå and Dingsøyr (2008b) to claim their assessment was similar, while Chen and Babar (2011) mentioned GRADE only to comment on result inconsistency—without conducting a full strength assessment across all required dimensions.

5.6 Making Recommendations in SE

In a tertiary review on the support that systematic reviews in SE provide for teaching about SE practice, Budgen et al. (2020) found that very few systematic literature reviews offer clear guidance to support effective decisions about technologies or practices. Our findings echo this: only a small number of secondary studies in SE provide clear recommendations to practitioners based on the evidence they aggregated and synthesized.

Given the nature of their findings, it is understandable that SMSs—one of the most frequent review types in our set—do not offer practitioner-oriented recommendations. However, both quantitative and qualitative systematic literature reviews have the potential to do so. Yet, only a few secondary studies in our dataset made recommendations—and all of them were qualitative systematic literature reviews.

There is still significant room for improvement, particularly in assessing the strength of such recommendations. For that, SE researchers must investigate outcomes of interest linked to interventions (e.g., practices, methods, tools) relevant to practitioners' daily work—an essential requirement for EBSE (Kitchenham et al. 2004). Assessing recommendation strength also requires understanding stakeholders' values and preferences regarding alternative interventions, as well as the resources needed to implement them. While SE has guidelines for evaluating resource use in intervention studies (Ali 2016), we still lack research and proposals addressing the other factors involved.

5.7 What else can we learn from Evidence-Based Medicine (and Perhaps what we will have to Figure Out Ourselves)?

When introducing EBSE, Kitchenham et al. (2004) discussed potential differences between EBSE and EBM, highlighting two main divergence points. First, SE involves a "skill factor": unlike many branches of medicine, SE "treatments" require practitioner expertise for proper execution. Second, there is the "lifecycle issue": most SE techniques affect only part of the development lifecycle, making it difficult to isolate their individual effects. This also complicates the assessment of their impact on final project outcomes, although immediate outcomes can still be measured. Kitchenham et al. (2004) proposed several actions to address these SE-specific challenges, such as developing experimental protocols to reduce bias, experimenting with isolated techniques, conducting large-scale empirical studies, and assessing the strength of evidence. This section discusses other differences we envision that may influence strength assessments and the derivation of recommendations for practitioners.

One such difference is structural and concerns the intended use of strength assessments. In medicine, these assessments inform clinical practice guidelines issued by international or national institutions, such as the World Health Organization, the Endocrine Society, or the American College of Cardiology. Physicians are expected to use these guidelines—together with their clinical expertise—to make treatment decisions. Increasingly, they are also expected to engage in shared decision-making, ensuring that patient values and preferences are considered. This raises important questions for SE: which institutions in our field have the authority to play a role analogous to medical societies? And do software professionals act more like physicians who use guidelines, or like patients who receive advice on options?

Another difference lies in the maturity of SE researchers to make strong recommendations in the GRADE sense. Perhaps we are not ready—or never will be—given the highly contextual nature of SE work (Dybå et al. 2012). If so, we might skip the step of deriving formal recommendations from strength assessments. Still, compiling all relevant information to support a potential recommendation may be valuable. In this case, practitioners and organizations—not professional societies—could determine what constitutes a strong or weak recommendation and use this information to tailor their own evidence-informed guidelines or policies to their own specific contexts.

Some efforts in SE have aimed to present evidence to practitioners. One such initiative was the Voice of Evidence column in IEEE Software. However, this column was later replaced by the Redirections column, which aimed to publicize surprising results in software engineering research investigations (Prikladnicki and Menzies 2018). Another still active initiative regards the publication of Evidence Briefings (EBs)—one-page summaries of

systematic literature review findings designed to facilitate communication with practitioners (Cartaxo et al. 2016). In a survey, practitioners responded positively to EBs, noting that the information was easy to locate, the format was clear, and the documents appeared trustworthy. However, many also felt that the EBs did not answer the practical questions they expected. GRADE tackles this by asking guideline developers to select outcomes that are critical to practitioners. EBs also lacked key information needed to decide whether to adopt an intervention, such as required resources—despite cost being a crucial factor (Ali 2016). The GRADE framework addresses such gaps by requiring an explicit analysis of benefits and harms, resource use, and implementation details.

Ultimately, choosing to adopt a SE method, tool, or practice is challenging. Like any decision involving human judgment, it is vulnerable to bias and noise. One effective strategy is to structure complex judgments (Kahneman et al. 2021), which involves three principles. First, break the decision into components—precisely what GRADE and GRADE-CERQual offer. Second, gather information about each component independently—something guideline panels can do. Third, delay judgment until all component information is available—perhaps a task for practitioners and organizations. In any case, the SE community can build on the structure of GRADE and GRADE-CERQual to help practitioners make better-informed decisions. Whether that requires more adaptation of these frameworks, use in the current format, or the development of a fully customized approach to the SE field is a topic for future research, as we discuss next.

5.8 A Call for Action: Using the Existing Strength Assessment Methods "as is", Adapting, or Developing New Ones?

Dybå and Dingsøyr (2008a) emphasized a critical limitation in the existing approaches for strength assessment, such as GRADE: these approaches are grounded in a hierarchy of evidence that prioritizes randomized experiments, even though such designs are not always feasible, and in some cases, observational studies can provide sufficiently strong evidence. This raises an important question: should we use these methods "as is", should we adapt them, or should we develop new ones from scratch?

Advocates of GRADE argue that it overcomes the simplistic view of traditional evidence hierarchies by offering a more nuanced framework in which study design is not the sole criterion (Djulbegovic and Guyatt 2017). As a result, it helps avoid "unwarranted confidence in RCTs [randomized controlled trials], as well as dogmatic decisions" (Djulbegovic and Guyatt 2017). If SE researchers accept that GRADE and GRADE-CERQual are appropriate for the SE context and choose to apply them "as is", the previous sections summarize key improvements required to support claiming the use of these methods:

- Choose the appropriate method based on the review type (Section 5.1);
- Use the most recent version of the method and apply all of its components (Section 5.2);
- Generate explicit, specific, and meaningful review findings (or organize them by outcome of interest) (Section 5.3);
- Do not equate quality assessment of primary studies with strength assessment of review findings (Section 5.4);
- Conduct specific strength assessments for each review finding (Section 5.5);
- Base recommendations on evidence and assess their strength (Section 5.6).

The list above is not intended as a set of recommendations or guidelines in the GRADE sense. Rather, it outlines steps to help move from the current state of strength assessments in SE toward the full adoption of the investigated methods, if the community chooses to follow that path. However, if SE researchers decide to adapt these methods, they must ask: what components should strength assessments consider? GRADE offers criteria for downgrading evidence from randomized controlled trials and for cautiously upgrading evidence from observational studies. Many primary studies in SE are observational. For example, Storey et al. (2020) showed that most papers published in ICSE (International Conference on Software Engineering) and EMSE (Empirical Software Engineering Journal) in 2017 used data strategies based on archival, generated, or simulated data. There is little reason to believe things have changed since then. Given this, do GRADE's upgrading components—large effect, dose-response gradient, or plausible confounding that would reduce the effect—apply to SE observational data?

GRADE-CERQual does not rely on study design as a starting point; instead, it uses only its set of prescribed assessment components. SE researchers may need to design a similar framework. What, for instance, would justify downgrading/upgrading evidence from data-driven studies, which are common in SE? Or from sources like expert panels, lessons learned, or feasibility studies? These questions remain open and highlight the need for further research—especially as SE reviews increasingly incorporate grey literature.

Additionally, the blind adoption or adaptation of existing methods is not the only path forward for strength assessments in SE. Our community may need to engage more broadly with questions about the kinds of evidence we produce and how they can be assessed in light of the inherent weaknesses and potential strengths of the research methods we employ (Stol and Fitzgerald 2018). It may also be necessary to reconsider whether starting from a specific study design and then downgrading or upgrading evidence is appropriate for our field. In any case, further discussion is essential if we want to fulfill the original goal of EBSE–serving as a means to integrate the best available research evidence with practical experience and human values in the decision making process of software development and maintenance, as first articulated by Kitchenham et al. (2004).

A final consideration is that strength assessments can benefit the SE community by providing synthesized and evaluated bodies of evidence that associations or committees may use when developing guidelines. At the same time, their adoption would increase the workload of systematic review authors, who would need to learn and apply additional procedures, and of reviewers, who would require training and more time to evaluate such studies. This highlights a trade-off between the effort required and the potential benefits to research quality and impact—a balance that the community will need to consider carefully.

6 Threats to Validity

Ampatzoglou et al. (2019) proposed addressing threats related to study selection, data, and research validity in review studies. Regarding **study selection**, there is a risk that the search process may have missed relevant papers. To mitigate this, we combined an automated search with snowballing based on a reference paper recognized in the SE research community. We piloted the automated search, refined the search string, and compared the results with a known set of papers from a previous tertiary review. Another threat comes from using the

forward snowballing of Dybå and Dingsøyr (2008a), as this step resulted in twelve out of the total 22 papers selected for the review. Since this source recommended GRADE explicitly, our results can be biased towards papers that adopted this method.

There is also a threat that we excluded relevant papers. To address this, we defined explicit inclusion and exclusion criteria and refined them during a pilot selection. We also applied a very inclusive first filter, selecting any SE secondary study at this stage. In the second, more restrictive filter, both authors independently selected a random subset of papers. We assessed inter-rater agreement, reaching a Cohen's Kappa of one. We also documented all selection decisions at this stage to ensure transparency in our rationale.

With respect to data validity, data extraction bias was a concern. We mitigated it through a pilot data extraction, which led to improvements in our extraction form. We decided to collect more information than initially planned and clarified the descriptions in our template. One author extracted data from all papers. To reduce associated risks, this author applied a test-retest approach: she extracted data from the same paper twice, with a four-month interval. The comparison revealed no critical differences, particularly in data used to answer the research questions (e.g., strength or recommendation assessments). The second author also validated the extraction for two papers and checked the corresponding data classifications.

We also considered the risk of bias in our classification schema. We developed the initial schema after the pilot extraction. Some categories, such as strength assessment methods and their dimensions, were defined from the outset and remained stable. Others emerged during data extraction and analysis, and we updated the schema accordingly.

Regarding **research validity**, we faced the threat that our process might be unreliable or unrepeatable. To mitigate this, we developed a research protocol to guide the study and documented the review execution in detail. We also made all extracted and analyzed data publicly available. Finally, there was a risk that our results might not be representative of the SE literature. We mitigated this by comparing our findings with those from a previous tertiary review (Budgen et al. 2020), noting disagreements only for specific recommendations in a few papers.

7 Final Considerations

The strength assessment of evidence is a cornerstone of evidence-based practice, as it enables the construction of trustworthy recommendations for practitioners based on systematic literature reviews. Although the SE research community was introduced to this concept nearly 20 years ago, our tertiary review shows that we still struggle to apply the existing approaches.

As a research community committed to impacting practice, we must decide whether to use the existing approaches "as is", adapt them, or develop new ones. If choosing to use them as they are, there is substantial room for improvement. We need to select the correct method based on the review type, adopt the most recent versions, and synthesize evidence into explicit, specific, and meaningful review findings—or organize them by outcome of interest. We must also conduct specific strength assessments for each finding and assess the strength of any resulting recommendations.

If we choose to adapt GRADE, the first step is to decide whether a hierarchy of evidence makes sense for the SE context, and if so, what is its structure. We must then identify and justify appropriate criteria for downgrading or upgrading evidence drawn from the various types of studies commonly conducted in our field.

Finally, GRADE or GRADE-CERQual are not perfect or universally optimal, and these approaches can be useful as starting points for discussing strength of evidence in SE. Given their extensive application in other fields, these frameworks offer structured, transparent criteria that SE researchers can adopt, adapt, or critique to develop more suitable practices in our own context.

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Data Availability All material generated during the current study is available at Figshare (https://doi.org/10.6084/m9.figshare.27448440.v1), as we describe here: Online Resource 1 presents the data extraction form for all secondary studies that we included in our tertiary study. Online Resource 2 presents the details of the selection process: all papers found, along with the filters and selection criteria that we applied.

Declarations

Competing interests The authors declare that they have no conflict of interest.

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